

1. What is buoyancy?

Correct:

- The tendency of an object to float in a fluid

Incorrect:

- The tendency of an object to sink in a fluid
- The resistance of an object to acceleration
- The force that pulls an object toward the ground

2. Which best describes buoyant force?

Correct:

- An upward force acting on a partially or fully submerged object

Incorrect:

- A downward force acting on an object in a fluid
- A horizontal force caused by fluid motion
- A force that exists only when an object sinks

3. Buoyant force acts in which direction relative to gravitational force?

Correct:

- Opposite to gravitational force

Incorrect:

- In the same direction as gravitational force
- Perpendicular to gravitational force
- At a 45° angle to gravitational force

4. Which principle states that buoyant force is equal to the weight of the displaced fluid?

Correct:

- Archimedes' principle

Incorrect:

- Pascal's principle
- Bernoulli's principle
- Newton's second law

5. Who is associated with the principle of buoyancy discussed in the module?

Correct:

- Archimedes

Incorrect:

- Isaac Newton
- Galileo Galilei
- Albert Einstein

6. According to Archimedes' principle, buoyant force is equal to the:

Correct:

- Weight of the displaced fluid

Incorrect:

- Mass of the object
- Weight of the object
- Volume of the object

7. An object may experience buoyant force when it is:

Correct:

- Partially or completely immersed in a fluid

Incorrect:

- Only completely submerged in a liquid
- Only partially submerged in a liquid
- Completely outside a fluid

8. In buoyancy, a fluid may refer to:

Correct:

- A liquid or a gas

Incorrect:

- A liquid only
- A gas only
- A solid only

9. For a completely submerged object, the volume of displaced fluid is:

Correct:

- Equal to the volume of the object

Incorrect:

- Smaller than the volume of the object
- Greater than the volume of the object
- Unrelated to the volume of the object

10. Which equation represents buoyant force?

Correct:

- $B = \rho g V$

Incorrect:

- $B = \rho V/g$
- $B = mg/V$
- $B = V/\rho g$

11. In $B = \rho g V$, what does ρ represent?

Correct:

- Density of the fluid

Incorrect:

- Buoyant force
- Gravitational acceleration
- Volume of displaced fluid

12. In $B = \rho gV$, what does V represent?

Correct:

- Volume of the displaced fluid

Incorrect:

- Velocity of the object
- Volume of the container
- Vertical acceleration

13. What value of gravitational acceleration is used in the module?

Correct:

- 9.80 m/s^2

Incorrect:

- 8.90 m/s^2
- 9.08 m/s^2
- 10.80 m/s^2

14. What is the SI unit of buoyant force?

Correct:

- Newton

Incorrect:

- Kilogram
- Cubic meter
- Kilogram per cubic meter

15. What is the SI unit of density?

Correct:

- kg/m^3

Incorrect:

- kg
- N
- M^3

16. What is the SI unit of displaced fluid volume?

Correct:

- m^3

Incorrect:

- m
- m^2
- kg/m^3

17. If fluid density increases while displaced volume stays constant, what happens to buoyant force?

Correct:

- It increases

Incorrect:

- It decreases
- It becomes zero
- It remains unchanged

18. If the volume of displaced fluid increases while density stays constant, what happens to buoyant force?

Correct:

- It increases

Incorrect:

- It decreases
- It becomes negative
- It remains constant

19. Two identical objects displace the same volume in two different fluids. Which object experiences greater buoyant force?

Correct:

- The object in the denser fluid

Incorrect:

- The object in the less dense fluid
- Both experience zero buoyant force
- Fluid density has no effect

20. Water has a density of 1000 kg/m^3 and brine has a density of 1030 kg/m^3 . For the same displaced volume, which produces greater buoyant force?

Correct:

- Brine

Incorrect:

- Water
- Both produce zero buoyant force
- Density does not matter

21. Ethyl alcohol has a density of 789 kg/m^3 while water has a density of 1000 kg/m^3 . For equal displaced volumes, which produces greater buoyant force?

Correct:

- Water

Incorrect:

- Ethyl alcohol
- Both always produce the same force
- Neither produces buoyant force

22. Why does an object appear to weigh less when submerged?

Correct:

- An upward buoyant force acts on the object

Incorrect:

- Its mass disappears
- Gravity stops acting on it
- Its density becomes zero

23. Which equation determines buoyant force from weight in air and weight in fluid?

Correct:

- $B = W_1 - W_2$

Incorrect:

- $B = W_1 + W_2$
- $B = W_2 - W_1$
- $B = W_1 W_2$

24. In $B = W_1 - W_2$, W_1 refers to:

Correct:

- Weight of the object in air

Incorrect:

- Weight of the fluid
- Weight of displaced fluid
- Weight of the object in fluid

25. In $B = W_1 - W_2$, W_2 refers to:

Correct:

- Weight of the object in fluid

Incorrect:

- Weight of the object in air
- Mass of the object
- Density of the fluid

26. An object weighs 10 N in air and 6 N in fluid. What is the buoyant force?

Correct:

- 4 N

Incorrect:

- 6 N
- 10 N
- 16 N

27. An object weighs 15 N in air and 11 N in fluid. What is its buoyant force?

Correct:

- 4 N

Incorrect:

- 2 N
- 11 N
- 26 N

28. A fluid has a density of 1000 kg/m^3 and the displaced volume is 0.002 m^3 . Using $g = 9.80 \text{ m/s}^2$, what is the buoyant force?

Correct:

- 19.6 N

Incorrect:

- 9.8 N
- 196 N
- 1960 N

29. A floating object is in equilibrium. What is the relationship between buoyant force and weight?

Correct:

- $B = W$

Incorrect:

- $B > W$
- $B < W$
- $B = 0$

30. A boat carrying the same load moves from fresh water to denser salt water. What happens to the volume of water it must displace?

Correct:

- It decreases

Incorrect:

- It increases
- It becomes zero
- It must double